

بسم الله الرحمن الرحيم



ISLAMIC UNIVERSITY OF GAZA

COMPUTER ENGINEERING DEPARTMENT

(Digital Aid Management and Distribution System)

Yasser Alaa Abu Amro	120200714
Feras Khaled Mustafa Alaqad	120200383
Ahmed Sami Ahmed Alabadla	120200956
Mohannad Ahmad Abdulhamid Alastal	120201239

Supervised By | **Dr.Hatem Hamad**

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ABSTRACT

The current aid distribution system faces challenges in coordination between aid providers and beneficiaries, leading to inefficient resource allocation and limited transparency. Many individuals in need struggle to find appropriate assistance, while institutions offering aid lack effective platforms to reach their target audiences. The Institution Connect Platform is a proposed comprehensive digital system designed to streamline the publication and management of aid-related announcements through four main interfaces: a public homepage, user page, institution page, and admin dashboard. Unlike existing platforms, it will provide real-time tracking to optimize resource allocation and include multilingual support to serve underserved communities.

The platform will employ a role-based architecture using modern web technologies, including React.js for frontend development, Python Django REST framework for backend services, and PostgreSQL for database management. The system will implement JWT-based authentication, admin verification processes, and comprehensive security measures to ensure credibility and transparency.

The Institution Connect Platform aims to address the gap between aid providers and recipients by offering a centralized, secure, and transparent ecosystem. Upon implementation, it is expected to improve efficiency in aid distribution coordination, enhance user experience through intuitive interfaces, and provide robust administrative controls to ensure content quality and platform security.

INTRODUCTION

This proposal outlines the plan for Graduation Project 2 to develop the Institution Connect Platform, a digital aid management and distribution system designed to connect aid-providing institutions with beneficiaries in a secure, transparent, and user-friendly manner. In Graduation Project 1, we defined the project idea, specified the technology stack (Python/Django REST Framework for the backend and React/Next.js for the frontend), designed the database, outlined base functionality, created a user journey map, and planned the administrative interface. No implementation was completed in Project 1.

In Graduation Project 2, we propose to fully develop and implement the platform, incorporating enhanced functionality such as a donation feature, additional interface pages, and multilingual support. The platform will be optimized for web accessibility across all devices, ensuring responsive interfaces that adapt seamlessly to desktops, tablets, and mobile browsers, providing a complete and operational system without the need for dedicated mobile applications.

❖ Work Completed in Graduation Project 1

Overview

Graduation Project 1 focused on conceptualizing the Institution Connect Platform and laying the groundwork for its development. The work was limited to planning and design, with no actual implementation, ensuring a clear vision and roadmap for Project 2.

Key Achievements

1. Project Idea and Objectives

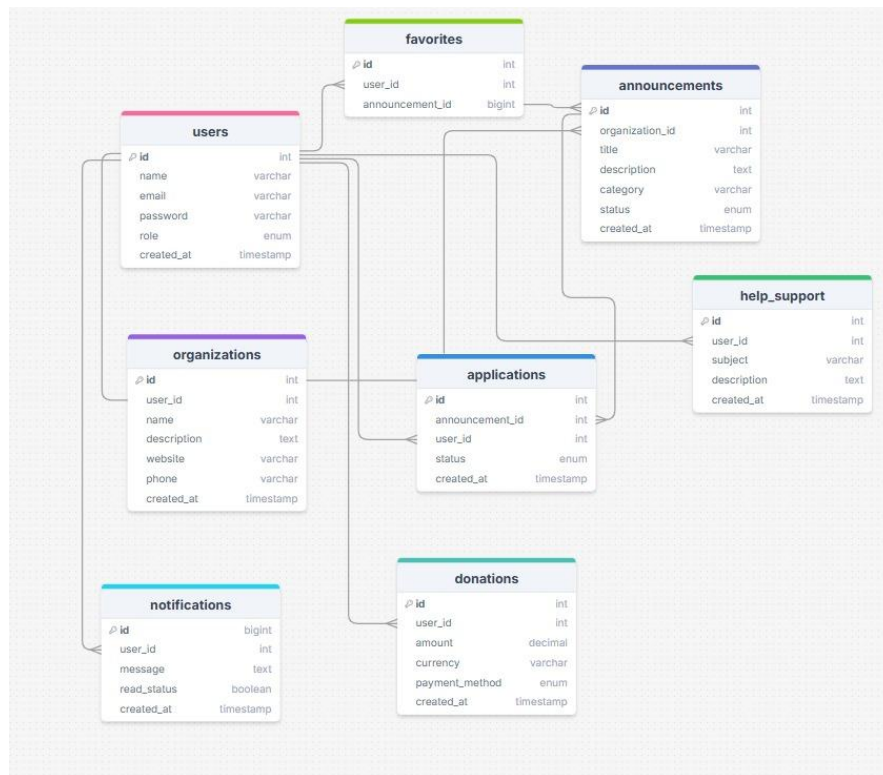
- Defined the platform's purpose: to address inefficiencies in aid distribution by creating a centralized, secure, and transparent system that connects institutions with individuals in need.
- Identified key challenges in existing systems: fragmented information, lack of verification, poor user experience, insufficient administrative controls, and security vulnerabilities.
- Established objectives:
 - Develop a user-friendly interface for accessing aid information.
 - Implement robust verification for institutions and content.
 - Provide administrative tools for platform management.
 - Enable efficient tracking and communication for aid distribution.

2. Technology Stack Selection

- Chose **Python/Django REST Framework** for the backend to build a scalable RESTful API for handling business logic and data operations.
- Selected **React/Next.js** for the frontend to create a dynamic, responsive, and SEO-friendly user interface.
- Adopted **Postgresql** as the database management system for efficient data storage and retrieval.
- Planned additional tools: Git for version control, Vercel for testing and preview, and Docker for deployment.

3. Database Design

- Designed a preliminary database schema with core entities.
- Created an Entity Relationship Diagram (ERD) to define relationships between entities.



4. Base Functionality Planning

- User Authentication: JWT-based authentication with role-based access control.
- Aid Announcement Management: Tools for institutions to create, edit, and submit announcements for approval.
- Search and Discovery: Search functionality with filtering by aid type and location.
- Request Submission: System for users to submit aid requests.
- Reporting System: Mechanism for users to report suspicious content.
- Planned security measures, including input validation and secure password storage.

5. User Journey Map

1. Public Visitor Journey

Goal: Explore the platform, view aid announcements, and register to access additional features.

- **Entry Point:** Accesses the Public Homepage via a web browser (responsive for desktop, tablet, and mobile).
- **Actions:**
 - Views platform overview, mission statement, and featured aid announcements.
 - Explores the “Donate Now” section and browses featured institutions.
 - Uses the search bar to filter aid announcements by type or location.
 - Registers as a user or institution via accessible forms (supports keyboard navigation, alt text).
 - Optionally logs in if already registered.

2. Registered User Journey

Goal: Search for aid opportunities, submit requests, manage donations, and report issues.

- **Entry Point:** Logs into the User Dashboard via the Public Homepage.
- **Actions:**
 - **Search and Discovery:**
 - Uses advanced search to find aid opportunities (filter by type, location, institution).
 - Saves preferred announcements to a favorites list.
 - **Aid Requests:**
 - Submits aid requests via a form with validation (e.g., required fields, format checks).
 - Tracks request status (pending, approved, rejected).
 - **Donations:**
 - Navigates to the donation page, selects an institution, and enters donation details.
 - Completes payment via a secure gateway (e.g., Stripe or Lahza).
 - Views donation history in a dedicated section.
 - **Reporting:**
 - Reports suspicious institutions or content via a dedicated reporting page.
 - Submits technical support queries or feedback through a help page with FAQs and contact forms.
 - **Notifications:**
 - Manages notification preferences for updates on requests or new announcements.

3. Institution Journey

Goal: Register, manage aid announcements, respond to requests, and track engagement.

- **Entry Point:** Registers via the Public Homepage and logs into the Institution Dashboard after admin verification.
- **Actions:**
 - **Registration and Verification:**
 - Submits institutional details for admin verification (e.g., credentials, contact info).
 - Awaits approval to access dashboard features.
 - **Aid Announcements:**
 - Creates, edits, and submits aid announcements for admin approval.
 - Manages published announcements (update, delete, or archive).
 - **Request Management:**
 - Views and responds to direct aid requests from users (approve, reject, or request more details).
 - **Donations:**
 - Tracks incoming donations (amount, donor, date) in the dashboard.
 - **Reporting:**
 - Reviews and addresses user-submitted reports forwarded by administrators.
 - **Analytics:**
 - Monitors announcement engagement (views, applications) and donation metrics.
 - **Notifications:**
 - Receives alerts for new requests, reports, or admin communications.

4. Administrator Journey

Goal: Manage platform content, verify institutions, monitor activity, and ensure security.

- **Entry Point:** Logs into the Admin Dashboard with elevated credentials.
- **Actions:**
 - **User and Institution Management:**
 - Reviews and verifies institution registrations (approve or reject based on credentials).
 - Manages user accounts (suspend, delete, or modify permissions).
 - Performs bulk actions (e.g., archive inactive accounts).
 - **Content Moderation:**
 - Approves or rejects aid announcements submitted by institutions.
 - Reviews user-submitted reports and forwards relevant ones to institutions.
 - **Donation Oversight:**
 - Monitors donation transactions for anomalies (e.g., fraud detection).
 - Generates reports for institutions on donation activities.
 - **Analytics and Monitoring:**
 - Views platform-wide analytics (user activity, announcement engagement, donation trends).
 - Uses predictive analytics to forecast aid demand.
 - Evaluates platform impact through reporting dashboards.
 - **System Configuration:**
 - Configures payment gateways, security policies, and notification settings.
 - Maintains audit logs for all administrative actions.
 - **Support:**
 - Responds to technical support queries from users and institutions.

6. Administrative Interface Planning

- Designed the admin page to include:
- Platform statistics and analytics.
- Tools for institution verification and content moderation.
- User management and security controls.
- Report handling and system configuration.

❖ Proposed Work for Graduation Project 2

Overview

Graduation Project 2 will focus on implementing the Institution Connect Platform based on the plans developed in Project 1. We will build the full system, enhance functionality with new features like donations. The project will follow an Agile methodology with iterative development and testing.

Objectives

1. Implement the core functionality planned in Project 1 to create a fully operational platform.
2. Integrate a donation feature to enable financial contributions to institutions.
3. Develop additional interface pages for reporting, help, and settings to enhance user experience.
4. Add multilingual support for broader accessibility.
5. Enhance administrative tools and analytics for effective platform management.
6. Conduct comprehensive testing and deployment to ensure reliability and usability.

Proposed Features and Enhancements

1. Donation Feature Implementation

- Objective: Enable users to make financial donations to verified institutions.
- Functional Requirements:
- Create a donation page in the user interface with a form to select an institution and enter an amount.
- Integrate a secure payment gateway (e.g., Stripe or Lahza) using Django REST Framework APIs.
- Develop an institution dashboard to track donations (amount, donor, date).
- Enable administrators to review donation transactions and forward related reports to institutions

2. Public Homepage Implementation

Build a responsive homepage with:

- A platform overview, mission statement, and featured aid announcements.
- A “Donate Now” section with a call-to-action button and featured institutions.
- Registration and login forms with accessibility features (keyboard navigation, alt text).

3. User Interface Implementation

Develop a user dashboard with:

- Advanced search and filtering for aid opportunities (by type, location, institution).
- A page for submitting aid requests with form validation.
- A dedicated reporting page for flagging suspicious institutions or personally identifiable information.
- A help and technical support page with FAQs and contact forms.
- A donation history section to track user contributions.
- Favorites and notification management features.

4. Institution Interface Implementation

Build an institution dashboard with:

- Tools for creating, editing, and managing aid announcements.
- A page to view and respond to direct aid requests from users.
- A section to review and address reports forwarded by administrators.
- Analytics for tracking announcement engagement and donation metrics.

5. Administrative Interface Implementation

Develop the admin dashboard with:

- Platform-wide analytics and monitoring tools.
- A donation management page to review all transactions.
- A reporting management page to handle user-submitted reports and forward them to institutions.
- A settings page for configuring payment gateways, security policies, and notifications.
- User and institution management tools with bulk actions and audit logs.

6. Multilingual Support

- Implement Arabic language support with right-to-left (RTL) interface using React/Next.js internationalization libraries.
- Enable content translation for announcements, interfaces, and user guides.
- Ensure cultural adaptation for local contexts.

8. Analytics and Reporting Enhancements

- Build advanced reporting dashboards for administrators and institutions using Django and React charting libraries.
- Implement predictive analytics to forecast aid demand and trends.
- Develop impact measurement tools to evaluate platform effectiveness.
- Provide performance optimization recommendations based on usage data.

Implementation Plan

Phase 1: Requirements Refinement and Setup (1 weeks)

- Validate stakeholder requirements for new features.
- Set up development environment (Django, React/Next.js, Postgres, Docker).
- Finalize database schema and interface designs .

Phase 2: Development (7 weeks)

- Implement backend APIs using Django REST Framework for core functionality and donations.
- Develop frontend interfaces with React/Next.js for all user roles.
- Integrate multilingual support and payment gateways and make server deployment for testing .
- Develop analytics and reporting dashboards.

Phase 3: Testing and Validation (2 weeks)

- Conduct unit and integration testing for APIs, interfaces.
- Perform security testing for donation transactions and live notification .
- Run user acceptance testing with real users.

Phase 4: Deployment and Finalization (2 weeks)

- Deploy the platform to a cloud (VPS or Dedicated server) production environment using Docker or another solution based on the feature requirements.
- Prepare a comprehensive final project report documenting the development process, features, and outcomes.
- Submit the final project report and deliverables to Dr. Hatem Hamad.

Conclusion

In Graduation Project 1, we successfully defined the concept, technology stack, database, base functionality, user journey map, and administrative interface for the Institution Connect Platform, setting a clear foundation for development. Graduation Project 2 will implement this vision, building a fully operational platform with enhanced features like donations, multilingual support, and mobile apps. We are committed to delivering a high-quality, scalable, and impactful solution under the supervision of Dr. Hatem Hamad, ensuring the platform meets the needs of all stakeholders and addresses critical gaps in aid distribution in Gaza.